I'M UP AND DOWN, AND ROUND AND ROUND

Circles Around Us

✓ **Circles in nature:**

- Early humans drew the sun as a circle in cave paintings. The cave paintings of **Gudahandi (Odisha)** show triangles, squares, circles and ovals, most likely copied from shapes seen in nature.
- Circular shapes in nature: ripples made by raindrops on water, the cross-section (the face seen when an object is cut straight across) of a plant stem, the inflorescence (the cluster of flowers on a plant) of a sunflower, the full moon, and the sun (clearly seen during a total solar eclipse).



(a) Ripples, stem and sunflower *(b) Moon; sun in a total eclipse*

Fig. 1: Circles seen in nature

✓ **The common property of all circles:**

- Every circle, big or small, has a **centre**.
- Every point of the circle is at the **same distance** from the centre.
- This single property is used to describe what a circle is (see 5.1).

✓ **Finding the centre of a paper circle (Jamuna's problem):**

- Jamuna has a circular piece of paper and wants its centre. Amina's tip works as follows.

1) Fold the circle so that its two halves overlap exactly, and open it. The crease (the line mark left by a fold) is a **diameter**.
2) Fold again in a different way so that the halves overlap, and open it. The second crease is another diameter.
3) Every diameter passes through the centre, so the point where the two creases cross is the **centre**.

Circle and Its Parts

✓ **Plane figures:**

- Circles, triangles and squares are always taken to be drawn on a flat sheet, that is, on a **two-dimensional plane**.

✓ **Meanings of important terms:**

- **Locus:** the set of all points that satisfy a given condition.
- **Circle:** the set (locus) of all points on a plane that are at the same distance from a fixed point on that plane.
- **Centre:** the fixed point from which all points of the circle are at equal distance.
- **Radius:** the distance from the centre to any point on the circle (also the segment joining them). All radii of a circle are equal.

Conceptual Learning

- **Chord:** a line segment joining any two points on the circle.
- **Angle subtended** (made (an angle) at a point) **by a chord at the centre:** the angle formed at the centre by joining the centre to the two ends of the chord. For chord CD with centre O, it is ∠COD.
- **Diameter:** a chord that passes through the centre. Diameter = 2 × radius.

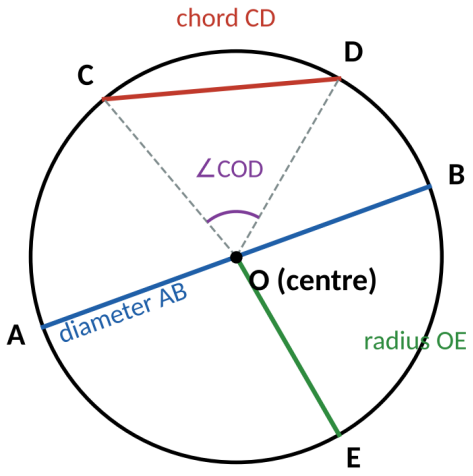


Fig. 2: Centre O, radius OE, chord CD, diameter AB, and ∠COD subtended by CD at O

✓ **Worked examples:**

- **Example 1:** a circle has centre A and B is on it. All points at distance AB from A form this circle, so its radius = AB.
- **Example 2:** a circle of radius 4 cm has diameter 2 × 4 = **8 cm**. A point 3 cm from the centre is **inside** the circle; a point 5 cm away is **outside**; a point exactly 4 cm away is **on** the circle.

Symmetries of a Circle

✓ **Rotational symmetry:**

- A rotating wheel looks the same at every moment. You cannot tell which point of it is touching the ground.
- **Complete rotational symmetry:** a circle turned about its centre through **any angle** looks exactly the same.

✓ **Reflection symmetry:**

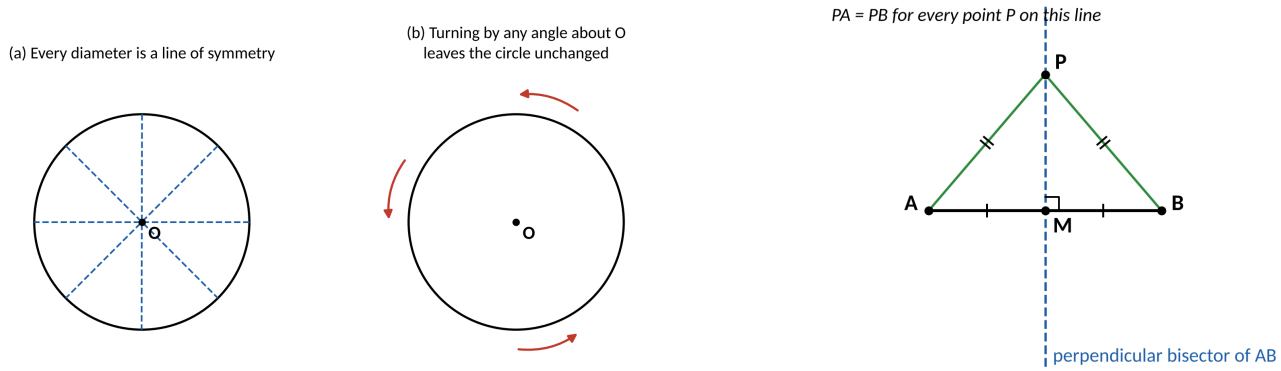
- Fold a paper circle so that the boundaries overlap, then open it. The crease is a **line of reflection symmetry**.
- The crease always passes through the centre, so it is a diameter.
- **Every diameter** is a line of reflection symmetry. So a circle has **infinitely many** lines of symmetry.

✓ **Perpendicular bisector as a locus:**

- **Perpendicular bisector of AB:** the line through the midpoint of AB that is at 90° to AB.
- Any point equidistant (at equal distance) from A and B lies on the perpendicular bisector of AB.
- Every point on the perpendicular bisector is equidistant from A and B.

Conceptual Learning

- So the **locus of points equidistant from A and B is the perpendicular bisector of AB**.



(a), (b) Symmetries of a circle *(c) PA = PB on the perpendicular bisector*

Fig. 3: Symmetries of a circle and the perpendicular bisector as a locus

✓ **Solved questions:**

Q1. What are the rotational symmetries of a square? How many lines of reflection symmetry does it have? What about a regular pentagon? A regular hexagon?

- Square: it looks the same after turns of 90°, 180°, 270° and 360°, so it has **4 rotational symmetries** (order 4) and **4 lines of symmetry** (2 diagonals and 2 lines joining midpoints of opposite sides).
- Regular pentagon: turns of 72°, 144°, 216°, 288°, 360° give **5 rotational symmetries**, and it has **5 lines of symmetry** (each from a corner to the midpoint of the opposite side).
- Regular hexagon: turns of 60°, 120°, ..., 360° give **6 rotational symmetries**, and it has **6 lines of symmetry** (3 through opposite corners, 3 through midpoints of opposite sides).
- A regular polygon with n sides has n rotational symmetries and n lines of symmetry.
- A circle has infinitely many of both.

Q2. What is the length of the longest chord in a circle of radius 5 units? Is there a smallest chord?

- The longest chord is the diameter = 2 × 5 = **10 units**.
- There is **no smallest chord**. A chord can be made shorter and shorter by moving it away from the centre, and its length gets as close to 0 as we like.

Q3. The locus of points at a given distance from a given point is a circle. What can we say about the locus of points equidistant from two given points A and B?

- Let M be the midpoint of AB and P any point on the perpendicular bisector.
- In △PMA and △PMB: AM = BM, ∠PMA = ∠PMB = 90°, PM is common.
- By SAS, △PMA ≅ △PMB, so PA = PB. So every point on the bisector is equidistant from A and B.
- Also, if PA = PB, then △PMA ≅ △PMB (SSS), so ∠PMA = ∠PMB = 90°, and P is on the bisector.
- So the locus is the **perpendicular bisector of AB**.

How Many Circles Pass Through Given Points?

✓ **Circles through one point:**

- Any point can be the centre and any length can be the radius, so **infinitely many** circles pass through one point.

Conceptual Learning

✓ **Circles through two points:**

- If a circle passes through A and B, its centre O satisfies OA = OB.
- So the centre lies on the **perpendicular bisector of AB**, and every point of this bisector can be a centre.
- So **infinitely many circles** pass through two points.
- **Smallest circle:** centre at the midpoint of AB; AB is its diameter, radius = **AB/2**.
- **No largest circle:** as the centre moves away from AB along the bisector, the radius keeps increasing without end.
- Larger circles look **less curved** (flatter) near A and B.

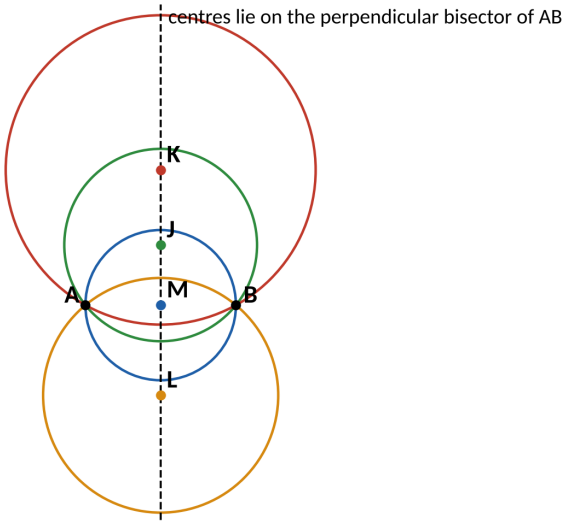


Fig. 4: Circles through A and B: centres M, J, K, L lie on the perpendicular bisector

✓ **Collinear points: no circle:**

- **Collinear** (lying on one straight line) **points:** points lying on one straight line.
- For collinear A, B, C, the perpendicular bisectors of AB and BC are both perpendicular to the same line, so they are **parallel** and never meet.
- So no point is equidistant from A, B and C, and **no circle passes through three collinear points**.
- Also, a line can cut a circle in at most **two** points.

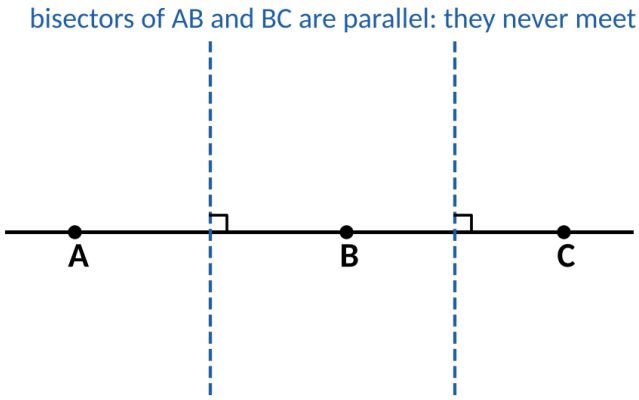


Fig. 5: Perpendicular bisectors of AB and BC for collinear points are parallel

✓ **Theorem 1: Circle through three non-collinear points:**

There is exactly one (unique) circle passing through three non-collinear points.

- **Proof:**
  1) Let the circle through A, B, C have centre O.
  2) OA = OB, so O lies on the perpendicular bisector of AB.
  3) OA = OC, so O lies on the perpendicular bisector of AC.
  4) A, B, C are not collinear, so these two bisectors are not parallel and meet at exactly **one** point. That point is O.
  5) The circle with centre O and radius OA passes through A, B and C. Since O is fixed, the circle is unique (only one).

✓ **Circumcircle and circumcentre:**

- **Circumcircle:** the circle through the three vertices of a triangle. It is said to **circumscribe** (draw around, touching all corners) the triangle.
- **Inscribed** (drawn inside with all corners on the circle) **triangle:** the triangle whose vertices lie on the circle.
- **Circumcentre:** the centre of the circumcircle; it is the meeting point of the perpendicular bisectors of the sides.
- To draw it: draw the perpendicular bisectors of any two sides, mark their meeting point O, and draw a circle with centre O and radius OA.

✓ **Position of the circumcentre:**

- **Acute-angled triangle:** circumcentre lies **inside** the triangle.
- **Obtuse-angled triangle:** circumcentre lies **outside** the triangle.
- **Right-angled triangle:** circumcentre lies at the **midpoint of the hypotenuse** (the side opposite the right angle).

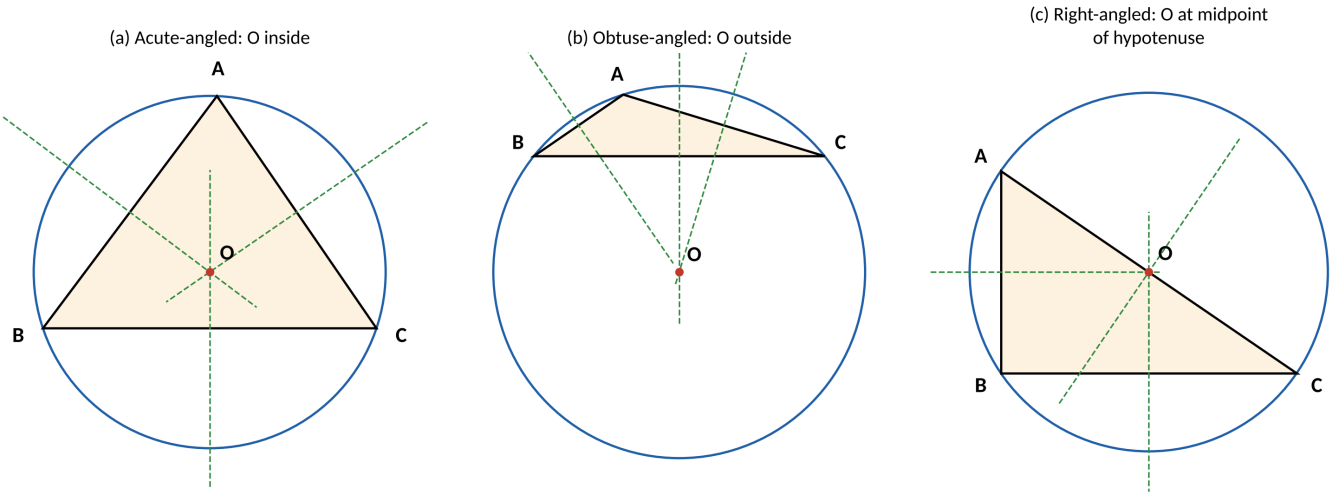


Fig. 6: Circumcircles of (a) acute, (b) obtuse and (c) right-angled triangles

✓ **Worked examples:**

- **Example 1:** a right triangle has legs 6 cm and 8 cm.
  - Hypotenuse = √(36 + 64) = 10 cm.
  - The circumcentre is the midpoint of the hypotenuse, so circumradius = **5 cm**.

Conceptual Learning

- **Example 2:** A and B are 8 cm apart.
  - The smallest circle through them has radius 4 cm.
  - A circle of radius 5 cm through them has its centre on the bisector at a distance $\sqrt{(25 - 16)} = 3$ **cm** from AB (on either side), so there are two such circles.

✓ **Solved questions:**

Q1. How many circles pass through two points on a plane?

➢ **Infinitely many.** Every point on the perpendicular bisector of the segment joining them is the centre of one such circle.

Q2. Are there circles of all possible radii passing through A and B? What is the radius of the smallest circle and of the largest circle through A and B?

➢ **No.** The radius cannot be less than AB/2, since the centre O must satisfy OA + OB ≥ AB, that is, 2r ≥ AB.

➢ Every radius from AB/2 upward is possible.

➢ Smallest radius = **AB/2** (centre at the midpoint of AB).

➢ There is **no largest circle**: the radius can be made as big as we like.

Q3. As you move away from segment AB along its perpendicular bisector, do the radii of the circles through A and B increase or decrease?

➢ If the centre is at distance d from the midpoint M, radius = $\sqrt{((AB/2)^2 + d^2)}$.

➢ As d increases, the radius **increases**.

Q4. As you go along the perpendicular bisector, will the circle drawn through A and B appear more curved or less curved?

➢ The radius keeps increasing, and a bigger circle bends less.

➢ So the circle looks **less curved**; near AB it becomes almost a straight line.

Q5. You are given two points A and B on a plane. How many squares can you draw with A and B on the boundary? How many squares can you draw with A and B as corners?

➢ **With A and B on the boundary: infinitely many**. A and B may lie on any sides, and squares of many sizes and positions are possible.

➢ **With A and B as corners: 3 squares.**

➢ Two squares have AB as a side, one on each side of the line AB.

➢ One square has AB as a diagonal.

Q6. (Exercise 5.1) Draw △ABC with AB = 5 cm, ∠A = 70° and ∠B = 60°. Draw its circumcircle. Is the centre inside or outside the triangle?

➢ ∠C = 180° − 70° − 60° = 50°.

➢ All angles (70°, 60°, 50°) are acute.

➢ Draw the perpendicular bisectors of AB and BC; they meet at O. The centre O lies **inside** the triangle.

Q7. (Exercise 5.1) Draw △ABC with AB = 5 cm, ∠A = 100°, AC = 4 cm. Draw the circumcircle. Is the centre inside or outside?

➢ ∠A = 100° is obtuse, so the triangle is obtuse-angled.

➢ The circumcentre lies **outside** the triangle, beyond the longest side BC.

Conceptual Learning

Q8. (Exercise 5.1) Draw △ABC with AB = 6 cm, BC = 7 cm and CA = 7 cm. Draw the circumcircle with centre O. Measure OA, OB, OC.

- Draw the triangle, then the perpendicular bisectors of AB and BC; they meet at O. The triangle is acute, so O is inside.
- OA = OB = OC (all radii).
- Check by calculation: R = (abc)/(4 × area)
- Height on AB = √(49 − 9) = √40 ≈ 6.32 cm, so area ≈ ½ × 6 × 6.32 ≈ 18.97 cm^2.
- R ≈ (6 × 7 × 7)/(4 × 18.97) = 294/75.9 ≈ **3.9 cm**. So OA = OB = OC ≈ 3.9 cm.

Q9. (Exercise 5.1) What is the least possible radius of a circle through two points A and B?

- **AB/2**, with centre at the midpoint of AB (AB is then a diameter).

Q10. A, B and C are three collinear points. Can you find a point P such that PA = PB = PC? What can you say about the perpendicular bisectors of AB and BC? Can a circle pass through collinear points? Can a line cut a circle in three distinct points?

- Both bisectors are perpendicular to the same line ABC, so they are **parallel** and do not meet.
- PA = PB needs P on the first bisector, and PB = PC needs P on the second; no point lies on both.
- So **no such P** exists.
- So **no circle** passes through three collinear points.
- Hence a line **cannot** cut a circle in three points; it meets a circle in at most two points.

Q11. The circumcircle of △ABC is drawn. Can other triangles congruent to △ABC share the same circumcircle?

- **Yes, infinitely many.**
- Rotate △ABC about the centre through any angle: the vertices stay on the circle and the new triangle is congruent (exactly equal in shape and size).
- Reflecting △ABC in any diameter also gives a congruent triangle on the same circle.

Chords and the Angles They Subtend

✓ **Thread on a wheel:**

- Tie a thread tightly between two points on a wheel. The thread acts as a chord and subtends (makes (an angle) at a point) an angle at the centre.
- On rotating the wheel, the thread becomes another chord of the **same length**, and the angle at the centre stays the **same**.

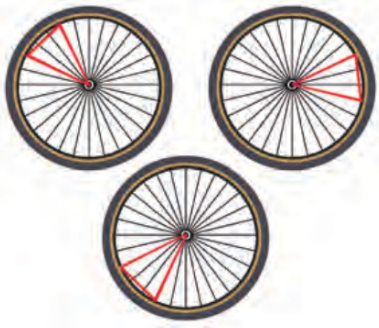


Fig. 7: A tight thread on a wheel acts as a chord

Conceptual Learning

✓ **Theorem 2: Equal chords subtend equal angles at the centre:**

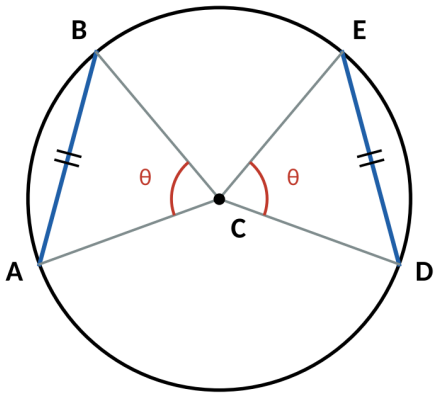
- **Given:** chords AB and DE of a circle with centre C, and AB = DE.
- **To prove:** ∠ACB = ∠DCE.

1) CA = CB = CD = CE (radii).
2) In △CAB and △CDE: CA = CD, CB = CE, AB = DE.
3) By **SSS**, △CAB ≅ △CDE.
4) So ∠ACB = ∠DCE (corresponding parts).

✓ **Theorem 3: Chords subtending equal angles at the centre are equal:**

- **Given:** ∠ACB = ∠DCE. **To prove:** AB = DE.
- Picture: stand at the centre with arms at a fixed angle and turn; the arm-ends trace chords that are all the same length.

1) CA = CD and CB = CE (radii).
2) ∠ACB = ∠DCE (given).
3) By **SAS**, △ACB ≅ △DCE.
4) So AB = DE.



AB = DE ⟺ ∠ACB = ∠DCE

Fig. 8: Theorems 2 and 3: equal chords and equal angles at the centre

✓ **Worked examples:**

- **Example 1:** chords PQ and RS of a circle are each 7 cm and ∠POQ = 50°. Then ∠ROS = **50°** (Theorem 2).
- **Example 2:** in a circle of radius 6 cm, chord AB subtends 60° at the centre O. △OAB has OA = OB and ∠O = 60°, so the other two angles are 60° each; it is equilateral. So AB = **6 cm**. Any chord subtending 60° at O also has length 6 cm (Theorem 3).

✓ **Solved questions:**

Q1. (Exercise 5.2) Show that the triangle formed by a chord and the centre of the circle is isosceles.

➢ Let AB be a chord and C the centre.

➢ CA and CB are radii, so CA = CB.

➢ Two sides are equal, so △CAB is **isosceles** (having two equal sides) with base AB.

Concept Learning

Q2. (Exercise 5.2) Show that if two such isosceles triangles have equal base length, they are congruent.

- Let △CAB and △CDE be formed by chords AB and DE with AB = DE.
- CA = CD and CB = CE (all radii).
- AB = DE (given)
- By **SSS**, △CAB ≅ △CDE.

Midpoints and Perpendicular Bisectors of Chords

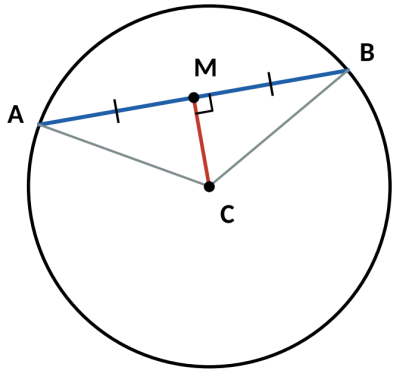
✓ **Theorem 4: Centre to midpoint of a chord is perpendicular to it:**

The line joining the centre of a circle to the midpoint of a chord is perpendicular to the chord.

1) Let AB be a chord, C the centre and M the midpoint of AB.
2) In △CMA and △CMB: CA = CB (radii), ∠A = ∠B (base angles of isosceles △CAB), AM = BM.
3) By **SAS**, △CMA ≅ △CMB, so ∠CMA = ∠CMB.
4) ∠CMA + ∠CMB = 180° (linear pair), so each angle is **90°**. Hence CM ⊥ AB.

✓ **Theorem 5: Perpendicular from the centre bisects the chord:**

The perpendicular from the centre of a circle to a chord bisects the chord. (This is the converse of Theorem 4; proof in Q1 below.)



CM ⊥ AB ⟺ AM = MB

Fig. 9: Theorems 4 and 5: CM ⊥ AB and AM = MB

✓ **Useful results:**

- The perpendicular bisector of any chord passes through the **centre**.
- In the right triangle CMA: $CA^2 = CM^2 + AM^2$, that is, **$r^2 = d^2 + (\text{half chord})^2$**, where d is the distance of the chord from the centre.
- So **chord length = $2\sqrt{(r^2 - d^2)}$**.

✓ **Worked examples:**

- **Example 1:** radius 5 cm, chord 8 cm. Half chord = 4 cm, so distance from centre = √(25 − 16) = **3 cm**.
- **Example 2:** radius 10 cm, chord at 6 cm from the centre. Half chord = √(100 − 36) = 8 cm, so chord = **16 cm**.

✓ **Solved questions:**

Q1. (Exercise 5.3) Explain why the perpendicular from the centre of a circle to a chord bisects the chord.

- Let CM ⊥ AB, with C the centre and M on AB.
- In △CMA and △CMB: ∠CMA = ∠CMB = 90°, CA = CB (radii, hypotenuses), CM common.
- By **RHS**, △CMA ≅ △CMB, so AM = BM. The chord is bisected.

Q2. (Exercise 5.3) An isosceles triangle ABC is inscribed in a circle, with AB = AC. Show that the altitude from A to BC passes through the centre.

- AB = AC, so A is equidistant from B and C; A lies on the perpendicular bisector of BC.
- The centre O is equidistant from B and C (radii), so O also lies on the perpendicular bisector of BC.
- In isosceles △ABC, the altitude from A is the perpendicular bisector of BC (it bisects the base).
- So the altitude from A passes through **O**.

Q3. (Exercise 5.3) Two parallel chords of lengths 6 cm and 8 cm are on opposite sides of the centre of a circle of radius 5 cm. Find the distance between the midpoints of the chords.

- Distance of 6 cm chord from centre = $\sqrt{(5^2 - 3^2)} = \sqrt{16} = 4$ cm.
- Distance of 8 cm chord from centre = $\sqrt{(5^2 - 4^2)} = \sqrt{9} = 3$ cm.
- Both midpoints lie on the same line through the centre (perpendicular to the chords), on opposite sides.
- Distance between midpoints = 4 + 3 = **7 cm**.

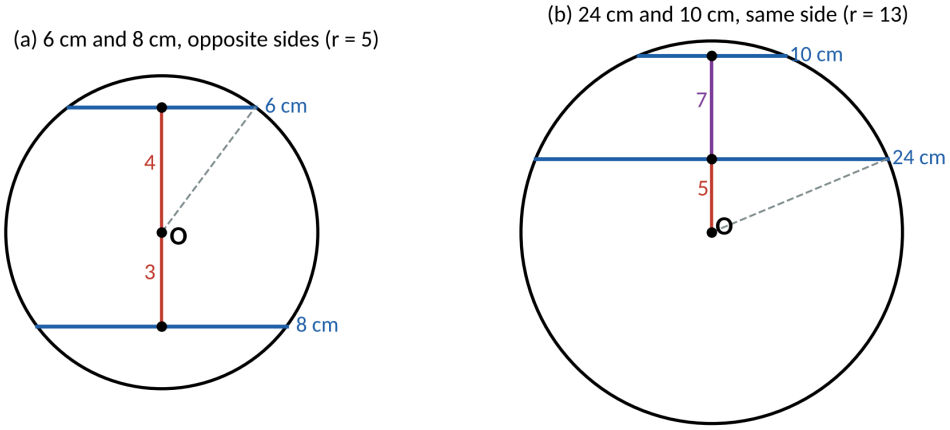


Fig. 10: Parallel chords: (a) on opposite sides of the centre; (b) on the same side

Q4. (End exercise 5) Prove that the perpendicular bisector of a chord passes through the centre of the circle.

- Let AB be a chord and O the centre.
- OA = OB (radii).
- O is equidistant from A and B, and the locus of such points is the perpendicular bisector of AB.
- So O lies on the **perpendicular bisector of AB**.

Q5. (End exercise 25) In a circle, chords CC and DD are perpendicular to a diameter AB. Prove that MM, joining the midpoints of CD and CD, is perpendicular to AB.

- AB passes through the centre and is ⊥ CC, so AB bisects (cuts into two equal parts) CC (Theorem 5)
- Let it meet CC at P: CP = PC.
- Similarly AB meets DD at Q with DQ = QD.

Concept Learning

- Take AB as the x-axis.
- Then C = (p, c), C = (p, −c), D = (q, d), D = (q, −d).
- M = ((p + q)/2, (c + d)/2) and M = ((p + q)/2, −(c + d)/2).
- M and M have the same x-coordinate, so MM is a vertical line, that is, **MM ⊥ AB**.
- (In other words, CD is the mirror image of CD in AB, so M is the mirror image of M.)

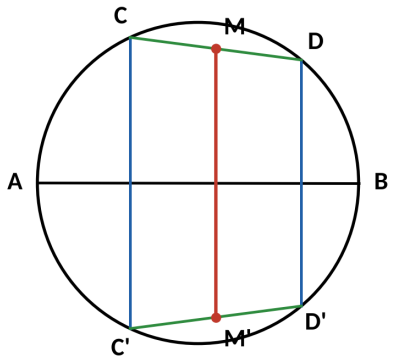


Fig. 11: Chords CC and DD ⊥ AB; MM ⊥ AB

Distance of Chords from the Centre

✓ **Distance of a chord from the centre:**

- **Distance of a chord from the centre:** the length of the perpendicular from the centre to the chord. It meets the chord at its midpoint (Theorem 5).

✓ **Paper folding check:**

1) Fold a paper circle inwards from the boundary and open it: the crease is a chord.
2) Fold again so that the two ends of this chord meet; open it.
3) The second crease cuts the chord into two equal parts at 90° and passes through the centre.
4) Draw another chord of the same length; its distance from the centre comes out the same.
5) Tracing the chord on tracing paper and rotating it about the centre gives other chords of the same length, each with the same distance from the centre.



Fig. 12: Folding a paper circle: (A) circle, (B) chord crease, (C) second fold through the centre

- Examples only suggest a result; a proof is needed to be sure.

✓ **Theorem 6: Equal chords are equidistant from the centre:**

- **Given:** AB = FG, centre C, E and H the midpoints of AB and FG. **To prove:** CE = CH.
- **Method 1 (congruent triangles):**

1) CA = CF and CB = CG (radii), AB = FG (given).

Conceptual Learning

2) By SSS, △CAB ≅ △CFG.
3) Congruent triangles have equal altitudes on equal bases, so CE = CH.

- **Method 2 (RHS):**
  1) AE = AB/2 = FG/2 = FH.
  2) ∠CEA = ∠CHF = 90° (Theorem 4).
  3) CA = CF (radii). By RHS, △CEA ≅ △CHF.
  4) So CE = CH.

✓ **Theorem 7: Chords equidistant from the centre are equal:**

If two chords are at the same distance from the centre, they are of equal length. (Converse of Theorem 6; proof in Q2 below.)

✓ **Theorem 8: The longer chord is nearer the centre:**

- **Given:** chords AB and DE with AB > DE; CF ⊥ AB and CG ⊥ DE. **To prove:** CF < CG.
  1) AC = CD (radii).
  2) By the Baudhāyana–Pythagoras theorem: $AC^2 = CF^2 + AF^2$ and $CD^2 = CG^2 + GD^2$.
  3) So $CF^2 + AF^2 = CG^2 + GD^2$.
  4) F and G are midpoints, so AF = AB/2 and GD = DE/2. AB > DE gives AF > GD, so $AF^2 > GD^2$.
  5) Hence $CF^2 < CG^2$, that is, **CF < CG**.

(a) Equal chords: CE = CH (b) Longer chord AB is nearer: CF < CG

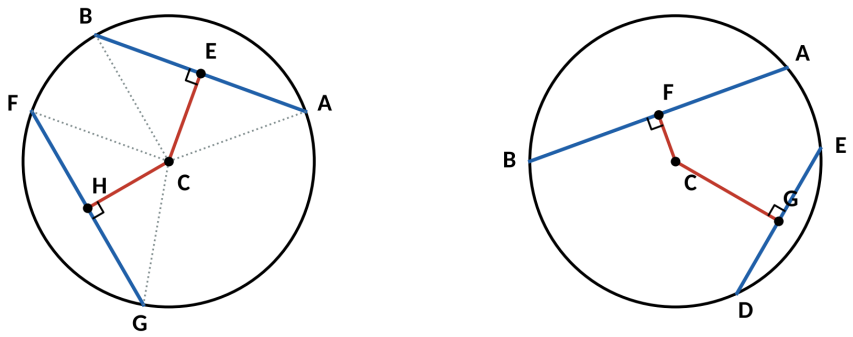


Fig. 13: (a) Theorem 6: equal chords, equal distances; (b) Theorem 8: longer chord is nearer

✓ **How chord length changes with distance:**

- The chord through the centre is at distance 0 and is the **longest chord (the diameter)**.
- As a chord moves away from the centre, it gets shorter. At distance r it shrinks to a single point (length 0).

✓ **Worked examples:**

- **Example 1:** radius 13 cm.
  - Chord at 5 cm: length $2\sqrt{(169 - 25)} = 24$ cm.
  - Chord at 12 cm: length $2\sqrt{(169 - 144)} = 10$ cm.
  - The longer chord (24 cm) is nearer, as Theorem 8 says.
- **Example 2 (table from drawing):** in a circle of radius 5 cm.

Conceptual Learning

Length of chord (cm)	10	8	6	4	0
Distance from centre (cm)	0	3	4	≈ 4.6	5

✓ **Solved questions:**

Q1. (Exercise 5.4) Use the Baudhāyana–Pythagoras theorem to show why Theorem 6 is true.

➢ Let AB = FG in a circle of radius r, with perpendiculars CE and CH from the centre.

➢ $CE^2 = r^2 - AE^2$ and $CH^2 = r^2 - FH^2$.

➢ AE = AB/2 = FG/2 = FH, so $CE^2 = CH^2$, and so **CE = CH**.

Q2. (Exercise 5.4) If CE ⊥ AB, CH ⊥ FG and CE = CH, show that AB = FG.

➢ In △CEA and △CHF: ∠CEA = ∠CHF = 90°.

➢ CA = CF (radii, hypotenuses), CE = CH (given).

➢ By RHS, △CEA ≅ △CHF, so AE = FH.

➢ E and H are midpoints (Theorem 5), so AB = 2AE = 2FH = FG.

Q3. (Exercise 5.4) Solve the previous question using the Baudhāyana–Pythagoras theorem.

➢ $AE^2 = r^2 - CE^2$ and $FH^2 = r^2 - CH^2$.

➢ CE = CH, so $AE^2 = FH^2$, that is, AE = FH.

➢ So AB = 2AE = 2FH = **FG**.

Q4. (Exercise 5.5) Find the length of a chord in a circle of radius 7 cm at a perpendicular distance of 6 cm from the centre.

➢ Half chord = $\sqrt{(7^2 - 6^2)} = \sqrt{13}$ cm.

➢ Chord = $2\sqrt{13} \approx$ **7.21 cm**.

Q5. (Exercise 5.5) If the distance of a chord from the centre is d and the radius is r, explain why the chord length is $2\sqrt{(r^2 - d^2)}$.

➢ The perpendicular from the centre meets the chord at its midpoint M (Theorem 5).

➢ In the right triangle formed by the centre, M and one end: $r^2 = d^2 + (\text{half chord})^2$.

➢ Half chord = $\sqrt{(r^2 - d^2)}$, so chord = **$2\sqrt{(r^2 - d^2)}$**.

Q6. (Exercise 5.5) If the distance of chord AB from the centre is twice that of chord CD, can we conclude that CD = 2AB? Explain why.

➢ **No.** Let CD be at distance d and AB at distance 2d.

➢ CD = $2\sqrt{(r^2 - d^2)}$ and AB = $2\sqrt{(r^2 - 4d^2)}$

➢ The lengths are not in a fixed ratio of 2.

➢ Example: r = 5, d = 2: CD = $2\sqrt{21} \approx 9.17$ cm, AB = $2\sqrt{9} = 6$ cm. 2AB = 12 ≠ 9.17.

➢ We can only say that CD is **longer** than AB (Theorem 8).

Q7. (End exercise 1) A chord is 5 cm away from the centre of a circle of radius 13 cm. Find its length.

➢ Half chord = $\sqrt{(13^2 - 5^2)} = \sqrt{144} = 12$ cm.

➢ Chord = 2 × 12 = **24 cm**.

Q8. (End exercise 3) The diameter of a circle is 26 cm. A chord of 24 cm is drawn. Find its distance from the centre.

Conceptual Learning

- Radius = 13 cm, half chord = 12 cm.
- Distance = $\sqrt{(169 - 144)} = \sqrt{25} =$ **5 cm**.

Q9. (End exercise 4) Radius 15 cm, distance of chord from centre 9 cm. Find the chord length.

- Half chord = $\sqrt{(225 - 81)} = \sqrt{144} = 12$ cm.
- Chord = **24 cm**.

Q10. (End exercise 9) A chord of length 16 cm is 6 cm from the centre. Find the radius.

- Half chord = 8 cm.
- $r = \sqrt{(8^2 + 6^2)} = \sqrt{100} =$ **10 cm**.

Q11. (End exercise 18) Two parallel chords of 10 cm and 24 cm lie on the same side of the centre, 7 cm apart. Find the radius.

- Let the 24 cm chord be at distance x from the centre.
- The shorter chord is farther, at distance x + 7.
- $r^2 = 12^2 + x^2$ and $r^2 = 5^2 + (x + 7)^2$.
- $144 + x^2 = 25 + x^2 + 14x + 49$, so $14x = 70$, $x = 5$ cm.
- $r^2 = 144 + 25 = 169$, so **r = 13 cm**. (Check: 10 cm chord at 12 cm: 25 + 144 = 169 ✓.)

Q12. (End exercise 13) Draw a circle in which a chord of 6 cm is at a distance of 3 cm from the centre.

- Radius = $\sqrt{(3^2 + 3^2)} = \sqrt{18} = 3\sqrt{2} \approx$ **4.24 cm**.
- Draw AB = 6 cm and its perpendicular bisector through the midpoint M.
- Mark O on the bisector with MO = 3 cm.
- Draw the circle with centre O and radius OA (≈ 4.24 cm)
- It passes through A and B, and AB is 3 cm from O.
- Link to the hint: MA = MB = MO = 3 cm, so $\angle AOB = 90°$.
- The circle is also the circumcircle of $\triangle ABX$, where X is the point where MO extended meets the circle.

Q13. (End exercise 16) Consider all chords of a fixed length l in a circle of radius r. What shape do their midpoints form?

- Every such chord is at the same distance $d = \sqrt{(r^2 - l^2/4)}$ from the centre (Theorem 6).
- The midpoint of each chord is the foot of this perpendicular, so it is at distance d from the centre.
- So the midpoints form a **circle with the same centre** (concentric) and radius $\sqrt{(r^2 - l^2/4)}$.

Q14. (End exercise 22) "No chord of a circle is longer than its diameter." Justify.

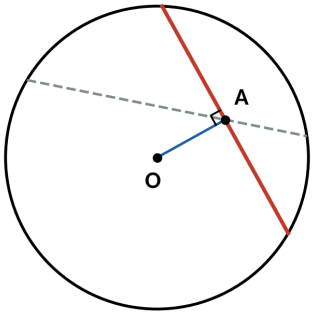
- Let AB be any chord and O the centre.
- In $\triangle OAB$ (or on the line if O is on AB): $AB \le OA + OB = r + r = 2r$ (triangle inequality).
- 2r is the diameter, so **AB ≤ diameter**, with equality only when AB passes through O.

Q15. (End exercise 23) A is a point inside a circle with centre O. Show that the shortest chord through A is the one perpendicular to OA.

- For any chord through A, let d be its distance from O. The perpendicular from O to the chord is at most OA, so $d \le OA$.
- $d = OA$ exactly when the chord is $\perp$ OA.
- Chord length = $2\sqrt{(r^2 - d^2)}$ is smallest when d is largest.

Conceptual Learning

➢ So the chord **perpendicular to OA** is the shortest.



red: shortest chord through A (⊥ OA)

Fig. 14: Shortest chord through A is ⊥ OA

Q16. (End exercise 12) Two equal chords AB and CD intersect at P. Show that the parts of one chord are equal to the corresponding parts of the other.

➢ Draw OM ⊥ AB and ON ⊥ CD. M and N are midpoints (Theorem 5).

➢ AB = CD, so OM = ON (Theorem 6).

➢ In right △OMP and △ONP: OP common, OM = ON. By RHS, △OMP ≅ △ONP, so MP = NP.

➢ AM = AB/2 = CD/2 = CN.

➢ So AP = AM + MP = CN + NP = CP, and PB = AB − AP = CD − CP = PD.

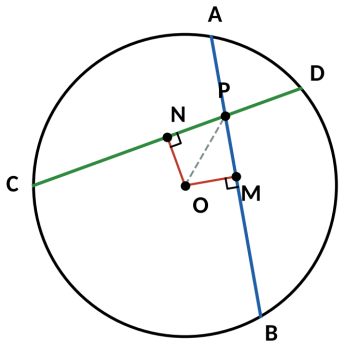


Fig. 15: Equal chords AB and CD meeting at P

Q17. (End exercise 17) Chords AB and AC of a circle with centre O are equal. Explain why the centre lies on the bisector of ∠BAC.

➢ In △OAB and △OAC: OB = OC (radii), OA common, AB = AC (given).

➢ By SSS, △OAB ≅ △OAC.

➢ So ∠OAB = ∠OAC. Hence AO bisects ∠BAC, and **O lies on the angle bisector**.

Q18. (End exercise 19) A regular hexagon is inscribed in a circle of radius r. Find its side and the distance of each side from the centre.

➢ Joining O to all 6 corners gives 6 equal triangles; each angle at O = 360° ÷ 6 = 60°.

➢ Each triangle has two sides r and the angle between them 60°, so it is equilateral.

Concept Learning

➢ **Side = r.**

➢ Distance of a side from O = altitude of an equilateral triangle of side r = $\sqrt{(r^2 - r^2/4)}$ = **$(\sqrt{3}/2)$ r.**

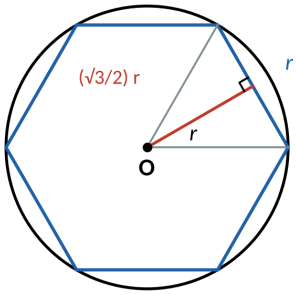


Fig. 16: Regular hexagon in a circle: side r, distance $(\sqrt{3}/2)$ r

Angles Subtended by an Arc

✓ **Arcs:**

- **Arc:** a connected part of a circle, fixed by two end points and the curve joining them along the circle.
- Two points A and B give two arcs. **Minor arc:** the smaller one. **Major arc:** the bigger one.
- **Angle subtended by an arc at the centre:** the angle swept by the radius as it moves from OA to OB along that arc.
- Minor arc: angle at centre **less than 180°**. Major arc: angle at centre **more than 180°** (a reflex angle). A semicircle: exactly 180°.

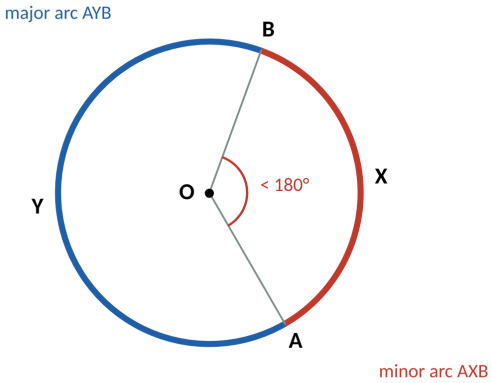


Fig. 17: Minor arc AXB (red) and major arc AYB (blue)

✓ **Angle subtended by an arc at a point on the circle:**

- The angle subtended by arc AXB at a point D on the circle but **not on the arc** is ∠ADB.
- This angle stays the **same** wherever D is taken on the remaining part of the circle.

✓ **Theorem 9: Angle at the centre is double the angle on the circle:**

The angle subtended by an arc at the centre is double the angle it subtends at any point on the remaining part of the circle.

- **Given:** arc AFB, centre C, D a point on the circle outside arc AFB. **To prove:** ∠ACB = 2∠ADB.

Conceptual Learning

- Join DC and extend it to meet the circle at E.
- **Case 1: E lies on arc AFB:**
  1) CB = CD (radii), so ∠CBD = ∠CDB.
  2) ∠BCE is an exterior angle of △BCD, so ∠BCE = ∠CBD + ∠CDB = 2∠BDC.
  3) Similarly CA = CD, so ∠ACE = ∠CAD + ∠CDA = 2∠CDA.
  4) Adding: ∠ACB = ∠ACE + ∠ECB = 2(∠CDA + ∠BDC) = **2∠ADB**.
- **Case 2: E lies outside arc AFB:**
  1) As above, ∠ACE = 2∠ADC and ∠BCE = 2∠BDC.
  2) Here ∠ACB = ∠ACE − ∠BCE and ∠ADB = ∠ADC − ∠BDC.
  3) So ∠ACB = 2(∠ADC − ∠BDC) = **2∠ADB**.

(a) Case 1: E lies on arc AFB

(b) Case 2: E lies outside arc AFB

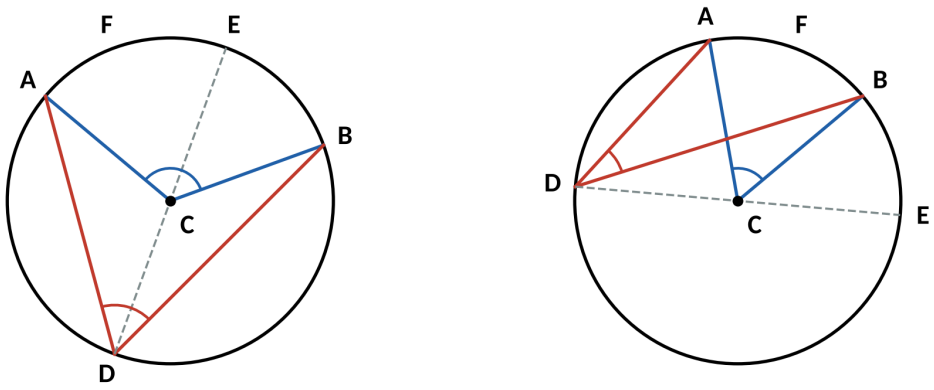


Fig. 18: Theorem 9: the two cases

✓ **Angles in the same segment are equal:**

- All points D, E, F on the circle outside arc AXB give ∠ADB = ∠AEB = ∠AFB = ½ ∠ACB.
- So **angles in the same segment of a circle are equal**. This property belongs only to the circle.

✓ **Corollary: Angle in a semicircle is 90°:**

(a) Angles in the same segment are equal (θ), and half the angle at the centre

(b) Angle in a semicircle = 90°

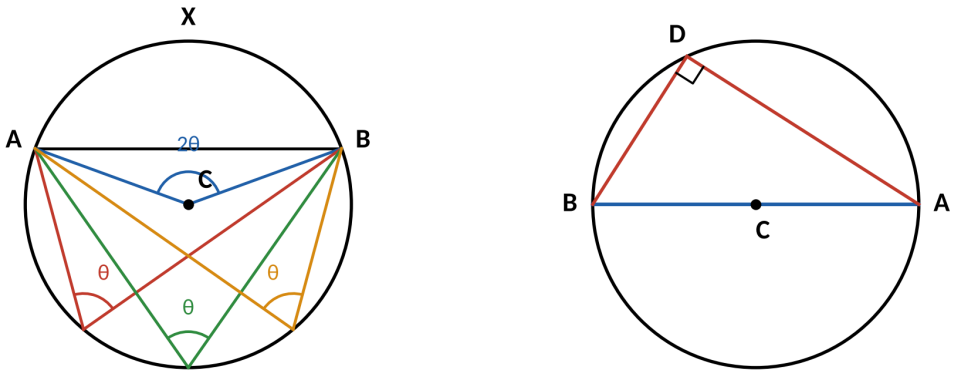


Fig. 19: (a) Equal angles in the same segment; (b) angle in a semicircle

- **Corollary:** a result that follows at once from a proved result.
- If AB is a diameter and D is any point on the circle, then **∠ADB = 90°**.
- Proof: the arc AB not containing D subtends a straight angle ∠ACB = 180° at the centre. By Theorem 9, ∠ADB = ½ × 180° = 90°.

✓ **Points inside, on and outside the circle:**

- For points **on** the circle (on the same side of AB), the angles subtended by AB are all equal.
- For a point **inside** the circle, the angle is **bigger**; for a point **outside**, it is **smaller**. Inside or outside points give different angles.

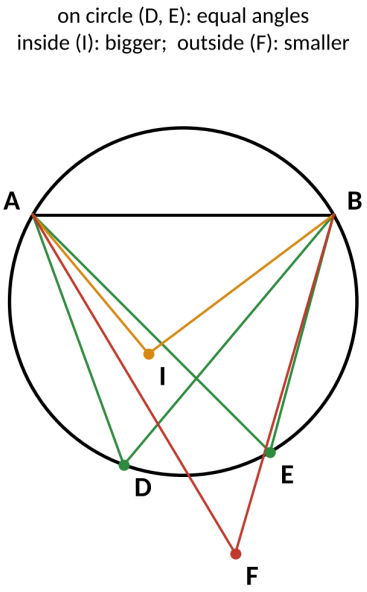


Fig. 20: Angles subtended by AB at points on, inside and outside the circle

✓ **Worked examples:**

- **Example 1:** an arc subtends 130° at the centre. At a point on the remaining part of the circle it subtends 130° ÷ 2 = **65°**.
- **Example 2:** P, Q, R, S lie on a circle, and ∠PRQ = 40°. S is on the same side of PQ as R, so ∠PSQ = **40°** (same segment), and ∠POQ = **80°** at the centre.
- **Example 3:** AB is a diameter and ∠BAC = 35°, with C on the circle. ∠ACB = 90°, so ∠ABC = 180° − 90° − 35° = **55°**.

✓ **Solved questions:**

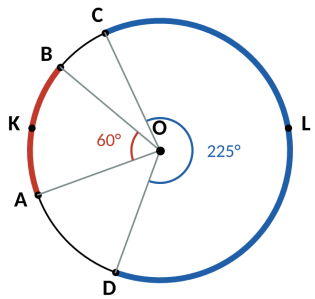


Fig. 21: Arc AKB (red) and arc CLD (blue)

Conceptual Learning

Q1. A circle with centre O has points A, B, C, D on it (figure below). Measure the angles subtended by arc AKB and arc CLD at O and state whether each is minor or major.

- Arc AKB subtends **60°** at O, less than 180°: **minor arc**.
- Arc CLD subtends **225°** at O, more than 180°: **major arc**.
- Rule: angle at centre < 180° gives a minor arc; > 180° gives a major arc.

Q2. (Exercise 5.6) In a circle with centre O, ∠AOB = 60°. If the radius is 12 cm, find chord AB.

- OA = OB = 12 cm, so ∠OAB = ∠OBA.
- ∠OAB + ∠OBA = 180° − 60° = 120°, so each is 60°.
- △OAB is equilateral, so **AB = 12 cm**.

Q3. (Exercise 5.6) A and B are two points on a circle with centre O. (i) Are there points X, Y on the circle, on the same side of AB, with ∠AXB ≠ ∠AYB? (ii) If ∠AXB = ∠AYB, must X and Y lie on the same side? (iii) If ∠AXB = ∠AYB and X, Y are not on the circle, does the circle through A, B and X also pass through Y?

- (i) **No.** X and Y lie in the same segment, so ∠AXB = ∠AYB always.
- (ii) **Usually yes.** Points on opposite sides give angles that add up to 180°.
- They are equal only if each is 90°, that is, when AB is a **diameter**; then X and Y can be on opposite sides.
- (iii) **Yes, if X and Y are on the same side of AB** (Theorem 10)
- If they are on opposite sides, the circle need not pass through Y.

Q4. (Exercise 5.6) Find x in the figure (ABCD on a circle, ∠ADC = 100°, x = ∠ABC).

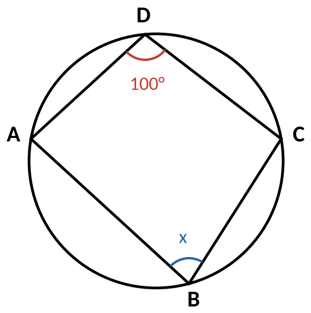


Fig. 22: Find x

- ABCD is a cyclic quadrilateral, so opposite angles add up to 180° (Theorem 11).
- x + 100° = 180°, so **x = 80°**.
- (Also by Theorem 9: arc ABC subtends 200° at the centre, the remaining arc 160°, and x = 160° ÷ 2 = 80°.)

Q5. (End exercise 2) An arc subtends 70° at the centre. Find the angle it subtends at a point on the circle.

- Angle on the circle = ½ × angle at centre.
- = 70° ÷ 2 = **35°** (at a point on the remaining part of the circle).

Q6. (End exercise 6) AB is a diameter and C is on the circle. Find ∠ACB and explain why.

- The arc AB not containing C is a semicircle and subtends 180° at the centre.
- By Theorem 9, ∠ACB = ½ × 180° = **90°**.

Q7. (End exercise 24) How would you use the figure to justify that the angle in a semicircle is 90°?

- O is the centre, BC is a diameter and A is on the circle, so OA = OB = OC.

Concept Learning

- In △OAB, OA = OB, so ∠OBA = ∠OAB = a.
- In △OAC, OA = OC, so ∠OCA = ∠OAC = b.
- Angle sum in △ABC: a + (a + b) + b = 180°, so 2(a + b) = 180°.
- a + b = 90°, that is, **∠BAC = 90°**.

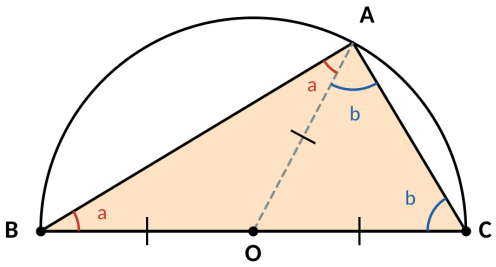


Fig. 23: Angle in a semicircle using isosceles triangles

Q8. (End exercise 20) Quadrilateral MNOP is inscribed in a circle and MN is a diameter. What can you say about ∠MOP and ∠MNP? Explain why.

- ∠MOP and ∠MNP are both subtended by chord MP at points O and N of the circle.
- N and O lie on the same arc between M and P, so they are in the same segment.
- So **∠MOP = ∠MNP**.
- Also, MN is a diameter, so ∠MON = ∠MPN = 90° (angle in a semicircle).

Concyclic Points and Cyclic Quadrilaterals

✓ **Meanings:**

- **Concyclic** (lying on the same circle) **points:** points that lie on the same circle.
- Any 3 non-collinear points are concyclic (Theorem 1). Four points may or may not be.
- **Cyclic quadrilateral:** a quadrilateral (4-gon) whose four vertices are concyclic.

✓ **Theorem 10: Test for four concyclic points:**

If segment AB subtends equal angles at two points C and D on the same side of AB, then A, B, C, D lie on a circle.

(a) D outside the circle (b) D inside the circle

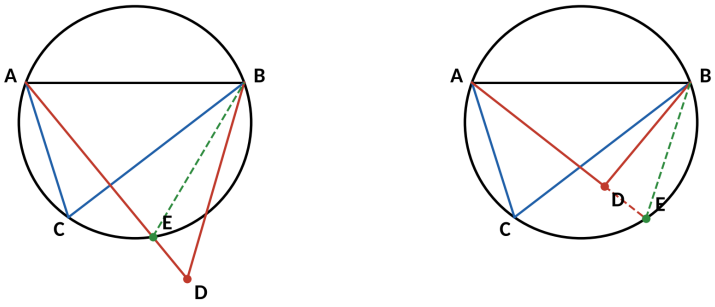


Fig. 24: Theorem 10: D cannot be outside or inside the circle

1) A, B, C are non-collinear, so a circle passes through them (Theorem 1). Suppose D is not on it.
2) **D outside:** AD cuts the circle at E. ∠AEB = ∠ACB (same segment). ∠AEB is an exterior angle of

△BED, so ∠AEB > ∠ADB = ∠ACB. This gives ∠ACB > ∠ACB, which is impossible.

3) **D inside:** extend AD to meet the circle at E. ∠ADB is an exterior angle of △BDE, so ∠ADB > ∠AEB = ∠ACB, again impossible.
4) So D lies **on** the circle, and A, B, C, D are concyclic.

✓ **Theorem 11: Opposite angles of a cyclic quadrilateral:**

The sum of either pair of opposite angles of a cyclic quadrilateral is 180°.

1) Let ABCD be cyclic with centre O.
2) A is outside arc BCD, so ∠BAD = ½ (reflex ∠BOD, the angle swept along arc BCD).
3) C is outside arc BAD, so ∠BCD = ½ (∠BOD swept along arc BAD).
4) The two angles at O together make a complete turn of 360°.
5) So ∠BAD + ∠BCD = ½ × 360° = **180°**. In the same way, ∠ABC + ∠ADC = 180°.

✓ **Theorem 12: Converse of Theorem 11:**

If a pair of opposite angles of a quadrilateral add up to 180°, the quadrilateral is cyclic.

1) Given ∠BAD + ∠BCD = 180°. Draw the circle through A, B, D. Suppose C is not on it.
2) **C outside:** let CD meet the circle at E. ABED is cyclic, so ∠BAD + ∠BED = 180° (Theorem 11).
3) Comparing with the given sum, ∠BED = ∠BCD.
4) But ∠BED is an exterior angle of △BEC, so ∠BED > ∠BCD. Contradiction.
5) **C inside:** extend DC to meet the circle at E; a similar argument gives ∠BCD > ∠BED = ∠BCD, again a contradiction.
6) So C lies on the circle, and ABCD is **cyclic**.

(a) Opposite angles of a cyclic quadrilateral (b) Proof of the converse: C outside

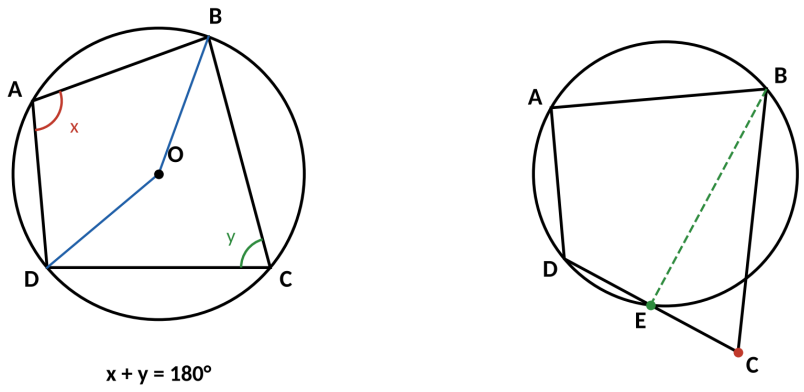


Fig. 25: (a) Theorem 11: x + y = 180°; (b) Theorem 12: C outside the circle

✓ **Exterior angle of a cyclic quadrilateral:**

- The exterior angle at any vertex equals the **interior opposite angle** (proved in Q5 below).
- A parallelogram that is cyclic must be a **rectangle** (Q7 below).

✓ **Worked examples:**

- **Example 1:** in cyclic ABCD, ∠A = 65° and ∠B = 120°. Then ∠C = 180° − 65° = **115°** and ∠D = 180° − 120° = **60°**. Check: 65 + 120 + 115 + 60 = 360° ✓.
- **Example 2:** a quadrilateral has angles 95°, 70°, 85°, 110° in order. 95° + 85° = 180°, so it is **cyclic** (Theorem 12).

Conceptual Learning

- **Example 3:** a quadrilateral has angles 90°, 80°, 100°, 90° in order. 90° + 100° = 190° ≠ 180°, so it is **not cyclic**.

✓ **Solved questions:**

Q1. A cyclic quadrilateral has ∠A = 80°, ∠B = 110°, ∠C = 100° and ∠D = 70°. Can it be drawn? Explain.

- ∠A + ∠C = 80° + 100° = 180° ✓.
- ∠B + ∠D = 110° + 70° = 180° ✓.
- Total = 360° ✓
- Both pairs of opposite angles add up to 180°, so **yes, it can be drawn** (Theorem 12).

Q2. (End exercise 7) In cyclic quadrilateral ABCD, ∠A = 75° and ∠B = 110°. Find ∠C and ∠D.

- ∠C = 180° − ∠A = 180° − 75° = **105°**.
- ∠D = 180° − ∠B = 180° − 110° = **70°**.

Q3. (End exercise 8) PQRS is inscribed in a circle, ∠P = (2x + 10)° and ∠R = (3x − 20)°. Find x, ∠P and ∠R.

- ∠P + ∠R = 180°: 2x + 10 + 3x − 20 = 180.
- 5x − 10 = 180, so 5x = 190, **x = 38**.
- ∠P = 2(38) + 10 = **86°**; ∠R = 3(38) − 20 = **94°**. Check: 86 + 94 = 180 ✓.

Q4. (End exercise 10) A cyclic quadrilateral has sides 5, 5, 12, 12 units. Find its area.

- Case 1 (equal sides next to each other, a kite ABCD with AB = AD = 5, CB = CD = 12): △ABC ≅ △ADC (SSS), so ∠B = ∠D. Being cyclic, ∠B + ∠D = 180°, so ∠B = ∠D = 90°.
- Area = 2 × ½ × 5 × 12 = **60 sq units**. (The diagonal AC = 13 is a diameter.)
- Case 2 (sides 5, 12, 5, 12 in order): opposite sides are equal, so it is a parallelogram; a cyclic parallelogram is a rectangle (Q7)
- Area = 5 × 12 = **60 sq units**.
- In both cases, area = **60 sq units**.

Q5. (End exercise 21) ABCD is cyclic and E is on CD produced. Explain why the exterior angle ∠ADE = ∠ABC.

- ∠ADE + ∠ADC = 180° (linear pair).
- ∠ABC + ∠ADC = 180° (Theorem 11).
- So **∠ADE = ∠ABC**: the exterior angle equals the interior opposite angle.

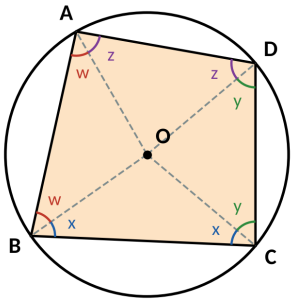


Fig. 26: Cyclic quadrilateral split into four isosceles triangles

Conceptual Learning

Q6. (End exercise 26) How would you use the figure to justify that opposite angles of a cyclic quadrilateral add up to 180°?

- Join centre O to A, B, C, D. OA = OB = OC = OD (radii), so △OAB, △OBC, △OCD, △ODA are isosceles.
- Their equal base angles are w, w (in △OAB), x, x (△OBC), y, y (△OCD), z, z (△ODA).
- Angles of ABCD: ∠A = w + z, ∠B = w + x, ∠C = x + y, ∠D = y + z.
- Sum of angles of ABCD = 2(w + x + y + z) = 360°, so w + x + y + z = 180°.
- ∠A + ∠C = w + z + x + y = **180°**, and ∠B + ∠D = w + x + y + z = **180°**.

Q7. (End exercise 14) Show that a rectangle is the only parallelogram that can be inscribed in a circle.

- In a parallelogram ABCD, opposite angles are equal: ∠A = ∠C.
- If it is cyclic, ∠A + ∠C = 180°.
- So 2∠A = 180°, ∠A = 90°.
- All angles are then 90°.
- So the parallelogram is a **rectangle**.

Q8. (End exercise 15) Show that if a rectangle is inscribed in a circle, its diagonals meet at the centre.

- ∠ABC = 90° and B is on the circle.
- By Theorem 9, the arc ADC subtends 2 × 90° = 180° at the centre, so AC is a **diameter**.
- Similarly, ∠BCD = 90° makes BD a diameter.
- Two diameters meet only at the centre.
- So the diagonals meet at the **centre**.

Q9. (End exercise 11) For a cyclic quadrilateral, how can we tell, without drawing the circumcircle, whether the centre lies inside or outside it?

- A side subtends an angle at each vertex not on it, e.g. side AB subtends ∠ACB at C.
- If this angle is **90°**, AB is a diameter and the centre lies **on** side AB.
- If it is **more than 90°**, the centre lies on the other side of AB from C, that is, **outside** the quadrilateral beyond AB.
- Best way: for each side, measure the angle it makes at one opposite vertex (∠ACB, ∠BDC, ∠CAD, ∠DBA)
- If all four are **acute**, the centre is **inside**; if any one is obtuse, it is outside.

Activities

✓ **Activity 1: Circles in nature:**

Aim:

- To list objects from nature that look like a circle.

Steps:

1) Look around at home, in a garden, and in the sky.
2) Note down every object whose outline or cut face looks circular.

Observations:

- Ripples on water, cut face of a tree trunk or stem, sunflower head, full moon, sun, eye pupil, orange slice, bird's nest opening, etc.

Conceptual Learning

- **Conclusion:**
  - Circles are very common in nature, which is why humans studied them from ancient times.

✓ **Activity 2: Folding a paper circle:**

- **Aim:**
  - To find the lines of symmetry and the centre of a paper circle, and to see how a chord relates to the centre.
- **Steps:**
  1) Draw a circle on paper and cut it out.
  2) Fold it so that the boundaries overlap; open it and look at the crease. Repeat with a different fold.
  3) Fold inwards from the boundary in any way to get a chord crease. Then fold so that the two ends of this chord meet.
  4) Measure the two parts of the chord, the angle between the creases, and the distance from the centre to the chord.
- **Observations:**
  - Each overlap-fold crease passes through the centre (a diameter); two such creases cross at the centre.
  - The second crease cuts the chord into two equal parts at 90° and passes through the centre.
- **Conclusion:**
  - Every diameter is a line of symmetry. The perpendicular from the centre bisects a chord, and the perpendicular bisector of a chord passes through the centre.

✓ **Activity 3: Chord length and distance from the centre:**

- **Aim:**
  - To find how the length of a chord is linked to its distance from the centre.
- **Steps:**
  1) Draw a circle (say radius 5 cm) and draw chords of different lengths.
  2) Drop a perpendicular from the centre to each chord and measure it.
  3) Record the length of each chord and its distance from the centre in a table.
- **Observations:**
  - Example readings: chord 10 cm at 0 cm, 8 cm at 3 cm, 6 cm at 4 cm, 4 cm at about 4.6 cm.
  - Chords of equal length have the same distance from the centre.
- **Conclusion:**
  - **The longer the chord, the closer it is to the centre.** The diameter is the longest chord.

✓ **Activity 4: Angles subtended by an arc:**

- **Aim:**
  - To compare the angle subtended by an arc at the centre with the angles it subtends at points on the rest of the circle.
- **Steps:**
  1) Draw a circle with centre O and a chord AB. Fix the arc AKB, with K between A and B.

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2) Measure ∠AOB subtended by arc AKB at the centre.
3) Take three points P, Q, R on the circle outside arc AKB. Measure ∠APB, ∠AQB and ∠ARB.
4) Repeat for a different arc AKB.

Observations:

- ∠APB = ∠AQB = ∠ARB each time.
- Each of them is half of ∠AOB (e.g. ∠AOB = 100° gives 50° at P, Q and R).

Conclusion:

- The angle subtended by an arc at the centre is **double** the angle it subtends at any point on the remaining part of the circle, and angles in the same segment are equal.

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